Knowledge Graph Completion Analysis

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1 Overview

This document analyses the completeness of our Music History knowledge graph (42,554 triples, 58 primary artists) and describes how RAG was used to resolve identified gaps. Completeness was measured using CM1–CM4 metrics [1], as taught in Week 10.

Gaps were identified through three methods: (1) CM1–CM4 metrics measuring schema coverage and property completeness, (2) SPARQL query results where sparse results indicated missing data, and (3) domain expert review of the TTL file. The team discussed the analysis after Meeting 3 and agreed to apply RAG to all 10 gaps.

2 Ontology Gaps

O1. `signedTo` lacks temporal scoping. The property links artists to all their labels simultaneously (229 triples) with no start/end dates. CQ17 produces cartesian products. RAG prompt P34 provided 226 temporal ranges (46% high confidence), demonstrating feasibility. Implementing temporal scoping requires RDF reification, which was documented as future work.

O2. `performedAt` sparsely populated. Initially only 5 triples from 3 artists. No structured source provides concert data. RAG prompt P35 identified 113 events across 43 artists. The use of programmatic ingestion and LLM API extraction brought the total to 131 triples.

O3. `collaboratedWith` modelled indirectly. MusicBrainz models collaborations as band/project entities rather than direct artist links. Starting from 76 triples, RAG prompt P36 identified 69 new direct pairs. Full LLM API extraction (58 artists) raised the total to 242 triples (3.2× increase).

O4. `founded` barely represented. Only 4 triples existed. RAG prompt P37 identified 47 new entries (labels, bands, foundations). Final count: 69 triples (17× increase).

O5. `hasMusicalPeriod` limited. Only 5 triples linked 3 artists to periods. RAG prompt P38 provided assignments for all 58 artists. Final count: 126 triples, covering every primary artist.

3 Instance Gaps

I1. Secondary artists lack data. Of 1,332 artist entities, only 58 are primary. The remaining 1,274 had minimal properties (`countryOfOrigin` 11.4%, `genre` 4.6%). RAG prompt P39 enriched 50 secondary artists (94% resolved). The pipeline's enrichment function was also extended to fetch genres, birth/death dates, and type, adding 216 triples for 102 artists.

I2. Umm Kulthum wrong entity. MusicBrainz search matched "UMM" (electronic project) instead of the Egyptian singer, causing a cascading failure: wrong entity → no Wikidata → no Wikipedia → no data. RAG prompt P40 provided comprehensive data but **hallucinated**

the MBID (returned 404). The correct MBID was found via programmatic API search. Fully resolved after adding an MBID override. Key insight: LLMs are unreliable for external identifiers; non-Latin scripts cause systemic entity resolution failures.

I3. LLM text extraction coverage. Initially 16/58 artists (298 triples, 28%). RAG prompt P41 demonstrated the approach for 5 artists, finding 44% of triples use text-unique predicates. The prompt template was then automated via Google Gemini API. **Final: 58/58 (100%), 897 triples.**

I4. Hildegard von Bingen missing genre. The only primary artist without genres. Medieval music predates modern classification, and MusicBrainz, Wikidata, and Discogs all failed. RAG prompt P42 provided 5 genres (medieval music, sacred music, plainchant), 3 periods, and 10 compositions. **Fully resolved.** Demonstrates RAG compensating for structured source blind spots.

I5. Cover composers lack data. 330 composer entities from cover detection had names/MBIDs but no biographical data. RAG prompt P43 enriched 50 composers (94% resolved) and surfaced 8 entity duplicates. The enrichment query was broadened to include composers, enabling automatic enrichment.

4 RAG Strategy and Results

Each gap was resolved using a 6-step RAG cycle following the "KG as SPARQL-based retrieval source" pattern from Week 11: (1) identify gap from the analysis above, (2) run a SPARQL query against the KG to retrieve relevant context, (3) verbalise retrieved triples into natural language, (4) prompt an LLM with the verbalised context and the gap question, (5) parse the LLM's JSON response into candidate triples, (6) ingest into the KG via programmatic entity linking. The ingestion script is integrated into the main pipeline as Step 14 and runs automatically after validation.

4.1 Concrete Example: Producer Gap-Filling

To illustrate the cycle, we demonstrate gap-filling for Michael Jackson's producer relationships. The SPARQL retrieval step fetched his 8 albums from the KG, including existing data (Quincy Jones produced Off the Wall, Thriller, and Bad). The RAG prompt provided this context and asked: "who produced each of these albums?" The LLM confirmed the 3 existing triples, provided 2 high-confidence new ones (Hal Davis for Got to Be There and Ben), and **flagged 3 as uncertain** (Dangerous, HIStory, Invincible — each had multiple co-producers). Only the 2 high-confidence triples were ingested; the 3 uncertain ones were documented for manual validation.

4.2 From RAG to Automation

A key outcome was the progression from manual RAG demonstration to pipeline automation. Several RAG findings were subsequently implemented as permanent improvements: I3 to LLM API (text extraction automated via Gemini, raising coverage from 16/58 to 58/58), I1/I5 to enrichment extension (fetch genres, dates, type for secondary artists), and data quality to smart filtering (Wikidata P31/P279* class hierarchy queries replacing manual blacklists).

4.3 RAG Results Summary

5 Completeness Metrics

CM1 Schema: 93% to 97% (29/30 properties populated). CM2 Primary: High to Higher (Country 98%, Genre 100%). CM2 All: Low to Improved (+216 triples from enrichment). CM4

Gap	Prompt	Result	Key Metric
O1 signedTo	P34	226 temporal ranges	46% high conf.
O2 performedAt	P35	113 events	5 → 131
O3 collaboratedWith	P36	69 direct pairs	76 → 242
O4 founded	P37	47 entries	4 → 69
O5 hasMusicalPeriod	P38	106 assignments	all 58 covered
I1 secondary artists	P39	50 enriched	94% resolved
I2 Umm Kulthum	P40	Full reconstruction	Entity fixed
I3 text extraction	P41	87 triples (demo)	28% → 100%
I4 Hildegard	P42	5 genres, 3 periods	Genre-less fixed
I5 cover composers	P43	50 enriched	94% resolved

Table 1: 10 RAG completion prompts. Total programmatically ingested: 494 triples.

Interlinking: Strong to Stronger (Umm Kulthum cross-refs fixed). Text extraction: 28% to 100% (897 triples via LLM API). Defined classes: 74 to 191 (+158%). Total triples: ∼38,000 to 42,554 (+12%).

6 Reflection

Three key insights emerged. First, **RAG is most valuable as a data quality auditor**: the KG vs Plain LLM comparisons (P18–P23) surfaced non-music genres, instrument duplicates, and range violations invisible to automated validation, leading to 6 pipeline fixes. Second, **LLMs are unreliable for external identifiers**: the Umm Kulthum case showed the LLM hallucinating a plausible MBID that returned 404. Non-Latin scripts cause cascading entity resolution failures. Third, **automated construction requires manual validation**: ontological correctness (Protégé) and semantic accuracy (domain expert review identifying 76 non-music genres and 31 non-music awards). The optimal approach is iterative: build → validate → review → fix → rebuild.

6.1 Future Work

1. Temporal scoping for `signedTo` via Wikidata qualifiers or RDF reification.
2. Concert/venue data from Setlist.fm API or MusicBrainz events.
3. Multilingual entity resolution to eliminate MBID overrides for non-Latin scripts.

References

[1] Zaveri, A., Rula, A., Maurino, A., Pietrobon, R., Lehmann, J., & Auer, S. (2016). Quality assessment for Linked Data: A survey. *Semantic Web*, 7(1), 63–93.